

AI Clinical Decision Support Lite

1. Executive Overview

The AI Clinical Decision Support Lite Hackathon is a specialized 5-day technical competition.

The goal is to design and implement a Retrieval-Augmented Generation (RAG) system that delivers evidence-based clinical recommendations

grounded strictly in official medical guidelines.

2. The Challenge

Large Language Models (LLMs) can generate fluent medical advice, but they often "hallucinate" or provide unsupported claims.

Participants must build a system that:

- Ingests official medical PDFs.
- Retrieves only relevant, high-quality evidence.
- Generates responses strictly based on that evidence.
- Provides transparent citations (Document Name, Section, Page).

3. Content & Technical Requirements

Teams must develop a 4-layer architecture:

1. **Document Ingestion Layer:** PDF parsing, section-aware chunking, and vector indexing.
2. **Retrieval Layer:** Semantic search optimization and transparent chunk display.
3. **Generation Layer:** Strict grounding prompts and structured citation formatting.
4. **Safety Layer:** Hallucination detection and refusal mechanisms for out-of-scope queries.

4. 5-Day Timeline

Day 1 — Research, Scope & Document Ingestion

Objective: Define clinical scope and build a reliable ingestion pipeline.

Tasks:

- Select 1–2 official guideline PDFs.
- Verify public accessibility and legal usability.
- Parse PDF content (structured extraction preferred).
- Implement section-aware chunking.
- Generate embeddings.
- Index chunks in vector database.
- Store metadata (document name, section, page number).

End-of-Day Outcome: Fully indexed guideline corpus and retrieval-ready vector database.

Day 2 — Retrieval Optimization

Objective: Achieve high-precision evidence retrieval.

Tasks:

- Implement top-k semantic search.
- Tune chunk size and overlap.
- Adjust k-value.
- Evaluate embedding model choice.
- Log retrieval scores.
- Display retrieved chunks before generation.

End-of-Day Outcome: Stable and explainable retrieval layer with evaluation logs.

Day 3 — Grounded Generation & Citation

Objective: Generate structured answers strictly grounded in retrieved evidence.

Tasks:

- Design strict grounding prompts.
- Prohibit external knowledge use.
- Format citations clearly (document name, section, page number).
- Structure response (recommendation, excerpt, citation).

- Implement refusal behavior when context is insufficient.

End-of-Day Outcome: Fully functional RAG pipeline with proper citation and refusal mechanism.

Day 4 — Safety, Guardrails & Internal Evaluation

Objective: Improve reliability and quantify performance.

Tasks:

- Add retrieval confidence threshold.
- Implement unsupported claim detection.
- Test with in-scope, ambiguous, and out-of-domain queries.
- Measure Retrieval Precision @ k.
- Measure citation accuracy.
- Evaluate faithfulness to retrieved text.

End-of-Day Outcome: Evaluated and stress-tested system with prepared safety demonstrations.

Day 5 — Final Presentation & Judge Evaluation

Objective: Demonstrate transparency, safety, and system rigor.

Presentation Requirements (15–30 minutes per team):

- Problem & Scope (clinical topic and guideline source).
- System Architecture (ingestion, retrieval, generation, safety).
- Live Demonstration with judge-provided query.
- Display retrieved chunks.
- Generated structured response with citation.
- Refusal case demonstration.

5. Selection Stages

1. **Initial Screening:** Verification of technical skills (Python/APIs) and team composition (2–4 members).
2. **Scope Approval (Day 1):** Mentors review and approve the chosen clinical guidelines for legal and technical suitability.
3. **Technical Milestone (Day 3):** Internal review of the functional RAG pipeline and citation accuracy.

4. **Final Evaluation (Day 5):** Live presentation and stress-testing by the judging panel.

6. Judging Criteria (100 Points)

- **Retrieval Precision (30 pts):** Accuracy of the evidence surfaced.
- **Answer Grounding & Citations (25 pts):** No hallucinations and precise source tracking.
- **Architecture Design (15 pts):** Technical clarity and modularity.
- **Evaluation Metrics (15 pts):** Quality of the team's internal testing methodology.
- **Clinical Safety (10 pts):** Proper handling of insufficient evidence.
- **UX & Live Demo (5 pts):** System usability and performance under pressure.

7. Staff Roles

- **Mentor:** Provides technical guidance on RAG logic and ensures teams stay within the medical scope.
- **Trainer:** Conducts workshops on prompt engineering, embeddings, and medical AI safety.
- **Operations:** Manages the event schedule, logistics, and communication between teams and judges.

8. Recommended Data Sources

U.S. Preventive Services Task Force (USPSTF) Recommendations

What it is: The gold standard for preventive medicine in the United States, covering screenings, counseling, and preventive medications.

Why it's perfect for RAG: The documents are strictly structured, publicly funded (no copyright hurdles for educational use), and feature explicit letter grades (A, B, C, D, I) for the strength of the evidence.

Where to find it: [USPSTF Recommendation Topics](#)

NICE Guidance Portals (UK)

What it is: The UK's National Institute for Health and Care Excellence provides comprehensive, evidence-based guidance for health and social care.

Why it's perfect for RAG: NICE documents are legendary for their rigorous formatting. They separate 'Recommendations' from 'Context' and 'Evidence,' making chunking and embedding highly predictable.

Where to find it: [NICE Published Guidance](#)

WHO Institutional Repository for Information Sharing (IRIS)

What it is: The official digital library of the World Health Organization's published material and clinical guidelines.

Why it's perfect for RAG: It offers a massive array of global health topics. The PDFs are open-access, clearly cited, and often come with executive summaries that are great for testing retrieval accuracy.

Where to find it: [WHO IRIS Portal](#)

European Society of Cardiology (ESC) Clinical Practice Guidelines

What it is: The leading authority on cardiovascular care in Europe.

Why it's perfect for RAG: Their guidelines are incredibly visual and structured, and they use a strict color-coded class of recommendation and level of evidence system. This is a great stress test to see if a RAG system can accurately extract the specific class of evidence.

Where to find it: [ESC Guidelines](#)

MAGICapp (Making GRADE the Irresistible Choice)

What it is: A global platform for authoring, publishing, and dynamically updating evidence-based guidelines. Many organizations (like BMJ Rapid Recommendations) publish here.

Why it's perfect for RAG: It uses the GRADE methodology for assessing evidence certainty. While it is a web platform, most guidelines have a clean, highly structured PDF export option suitable for text chunking.

Where to find it: [MAGICapp Platform](#)